# Sec. 2.2 Separable equations

$$\frac{dy}{dx} = f(x,y) \quad \text{or} \quad \frac{dy}{dx} - f(x,y) = 0$$

$$\underline{M(x,y) + N(x,y)\frac{dy}{dx} = 0}$$

The differential eq. $\downarrow$ is separable

$$\Leftarrow \begin{cases} M(x,y) = M(x) \\ N(x,y) = N(y) \end{cases}$$

$$\text{or} \quad \underline{M(x) + N(x)\frac{dy}{dx} = 0}$$

Find $H_1(x)$ s.t. $H_1'(x) = M(x)$,

find $H_2(x)$ s.t. $H_2'(y) = N(y)$

$$M(x) + N(x)\frac{dy}{dx} = 0$$

$$\Rightarrow H_1'(x) + H_2'(y)\frac{dy}{dx} = 0$$

$$\Leftrightarrow \frac{dH_1}{dx} + \underbrace{\frac{dH_2}{dy}\frac{dy}{dx}}_{\text{chain rule} = \frac{d}{dx}H_2(y(x))} = 0$$

$$\Leftrightarrow \frac{d}{dx}\left(H_1(x) + H_2(y(x))\right) = 0$$

Thus $\underline{H_1(x) + H_2(y(x)) = C}$

This eq. defines the solution $y = y(x)$ implicitly

E.g.: $\frac{dy}{dx} = -\frac{x}{y}$

$\Leftrightarrow x + y\frac{dy}{dx} = 0$

① or $\frac{d}{dx}\left(\frac{x^2}{2} + \frac{y^2(x)}{2}\right) = 0$

$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} = C$

or $x^2 + y^2 = 2C$

$dx \; y\frac{dy}{dx} = -\frac{x}{y}\, y\, dx$

② $\int y\,dy = -\int x\,dx$

$\frac{y^2}{2} = -\frac{x^2}{2} + C$

$M(x) + N(y)\frac{dy}{dx} = 0$

$M(x)\,dx + N(y)\,dy = 0$

$\int H_1'(x)\,dx = -\int H_2'(y)\,dy$

$H_1(x) = -H_2(y) + C$

$H_1(x) + H_2(x) = C$

$$\begin{cases} M(x) + N(y)\dfrac{dy}{dx} = 0 \\ y(x_0) = y_0 \end{cases}$$

$$H_1(x) + H_2(y(x)) = C$$

$$\Leftrightarrow H_1(x_0) + H_2(y_0) = C$$

$$\text{or} \begin{cases} H_1(x) + H_2(y) = H_1(y_0) + H_2(y_0) \\ H_1(x) - H_1(x_0) + H_2(y) - H_2(y_0) = 0 \end{cases}$$

$$\int_{x_0}^{x} H_1'(s)\,ds + \int_{y_0}^{y} H_2'(s)\,ds = 0$$

$$\int_{x_0}^{x} M(s)\,ds + \int_{y_0}^{y} N(s)\,ds = 0$$

is the solution to $\begin{cases} M(x) + N(y)\dfrac{dy}{dx} = 0 \\ y(x_0) = y_0 \end{cases}$

Homogeneous Equations

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

① $\frac{dy}{dx} = \frac{y-2x}{y+5x}$

E.g.: Solve $\frac{dy}{dx} = \frac{4x-x^3}{4+y^3}$

$$\int (4+y^3)\,dy = \int (4x-x^3)\,dx$$

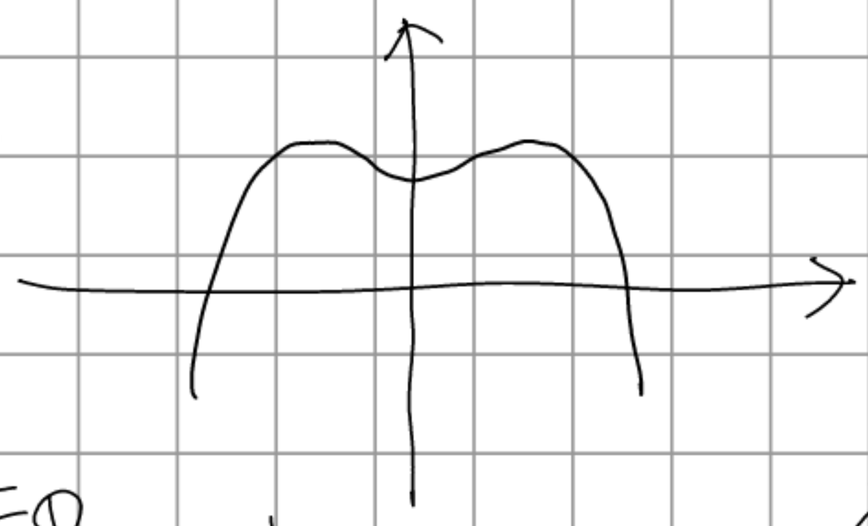
$$4y + \frac{y^4}{4} = 2x^2 - \frac{x^4}{4} + C$$

找：过 $(0,1)$ 点的 sol.

$$4 + \frac{1}{4} = C$$

$$4y + \frac{y^4}{4} = 2x^2 - \frac{x^4}{4} + \frac{17}{4}$$

画图：



② $4 + y^3 = 0$

$\Rightarrow y = (-4)^{\frac{1}{3}} \simeq -1.6$ for $x_0 = 3.35$

The sol. is given implicitly by

$$4y + \frac{y^4}{4} = 2x^2 - \frac{x^4}{4} = \frac{17}{4} \text{ on } (-3.35, 3.35)$$

interval of existence

Solve $\frac{dy}{dx} = \frac{3x^2+4x+2}{2(y-1)}$

and determine the $y(0) = -1$

(2) interval in which the sol. exists

$$\int 2(y-1)\,dy = \int (3x^2+4x+2)\,dx$$

$$y^2 - 2y = x^3 + 2x^2 + 2x + C$$

$$(-1)^2 - 2\cdot(-1) = 0 + C \Rightarrow \underline{C=3}$$

$$y^2 - 2y = x^3 + 2x^2 + 2x + 3$$

$$y = \frac{2 \pm \sqrt{4 + 4(x^3+2x^2+2x+3)}}{2}$$

$$= 1 \pm \sqrt{x^3+2x^2+2x+3+1} \quad (=+4)$$

$$y(0) = 1 \pm \sqrt{4} = 1 \pm 2 = \cancel{3} \text{ or } -1$$

$$\Rightarrow y(x) = 1 - \sqrt{\quad}$$

Need: $\begin{cases} x^3+2x^2+2x+4 > 0 \\ y \neq 1 \end{cases}$

$$\rightarrow x(x^2+2) + 2(x^2+2) > 0$$

$$(x^2+2)(x+2) > 0$$

$(x^2+2 > 0)$ ← $0 \in (-2, \infty)$

$$\Rightarrow x + 2 > 0 \Rightarrow x > -2$$

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Solve $\frac{dy(x)}{dx} = f\left(\frac{y(x)}{x}\right)$

Last time: $M(x) + N(y)\frac{dy}{dx} = 0$

Set $u = \frac{y}{x}$, $y = ux = u(x)\cdot x$

$f\left(\frac{y}{x}\right) = f(u)$

$\frac{dy}{dx} = \frac{du}{dx}x + u$

$\frac{du}{dx}x + u = f(u)$

or $x\frac{du}{dx} = f(u) - u$

? $\int \frac{dx}{x} = \int \frac{du}{f(u)-u}$

$\ln|x| = \int \frac{du}{f(u)-u}$

Recover $u(x)$ and then $y(x) = x\,u(x)$

Solve $\frac{dy}{dx} = \frac{x+3y}{x-y} = \frac{1+3\frac{y}{x}}{1-\frac{y}{x}}$

Let $u = \frac{y}{x}$, $y = ux$

$\frac{dy}{dx} = u'x + u = \frac{1+3u}{1-u}$ ↓ ?

$u'x = \frac{1+3u}{1-u} - u$

$= \frac{u^2+2u+1}{1-u}$

$= \frac{(u+1)^2}{1-u}$ (Test $u=-1$ : $y=-x$ is a sol. to the diff. eq.)

$x\frac{du}{dx} = \frac{u^2+2u+1}{1-u}$

$\Rightarrow \int \frac{dx}{x} = \int \frac{1-u}{u^2+2u+1}\,du$

$= \int \frac{du}{(u+1)^2} - \int \frac{u}{u^2+2u+1}\,du$

$= -\frac{1}{1+u} - \int \frac{u+1-1}{u^2+2u+1}\,du$

Let $z = u^2+2u+1$
$\Rightarrow dz = (2u+2)\,du$

$= \frac{1}{1+u} + \int \frac{du}{(1+u)^2} - \frac{1}{2}\int \frac{2u+2}{u^2+2u+1}\,du$

$= -\frac{2}{1+u} - \frac{1}{2}\int \frac{dz}{z}$

$= -\frac{2}{1+u} - \frac{1}{2}\ln|z|$

$= -\frac{2}{1+u} - \frac{1}{2}\ln(u+1)^2$

$= -\frac{2}{1+u} - \ln|u+1|$

$\left(\ln\frac{a}{b} = \ln a - \ln b\right)$

② $\ln|x| = -\frac{2}{1+\frac{y}{x}} - \ln\left|\frac{y}{x}+1\right| + C$

$\Rightarrow \ln|x| = -\frac{2x}{y+x} - \ln|y+x| + \ln|x| + C$

$\Rightarrow \frac{2x}{y+x} + \ln|y+x| = C$

② $y = -x \longrightarrow$ Sol.

Bernoulli: $y' + p(t)y = q(t)y^n$,

$n \neq 0, 1$,

$n = 2, 3, 4, \cdots$,

$p(t), q(t)$ one cont.

$n=0$: $y' + p(t)y = q(t)$ ($\rightarrow$ Sec. 2.1)

$n=1$: $y' + (p(t)+q(t))y = 0$

$\Rightarrow y(t) = Ce^{-\int [p(t)-q(t)]\,dt}$

$n \neq 0, 1$: $y^{-n}\frac{dy}{dt} + p y^{1-n} = q$

Set $u = y^{1-n}$

$\Rightarrow \frac{du}{dt} = (1-n)y^{-n}\frac{dy}{dt}$

$\frac{1}{1-n}\frac{du}{dt} + p(t)u = q(t)$ ($\rightarrow$ Sec. 2.1)

$\rightarrow$ Solve for $u$ and then

solve for $y$ from $u = y^{1-n}$

Riccati's Equation

$$\frac{dy}{dt} = q_1(t) + q_2(t)\,y + q_3(t)\,y^2$$

If $y_1(t)$ is a sol.

then the general sol. can be found

by the substitution

$$y(t) = y_1(t) + \frac{1}{u(t)}$$

$\Rightarrow$ linear diff. eq. in $u$

Sec. 2.6

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0$$

Assume that $\exists \psi(x,y)$, s.t. $\psi_x(x,y) = M(x,y)$

& $\psi_y(x,y) = N(x,y)$

$$\begin{cases} \psi_x + \psi_y \frac{dy}{dx} = 0 \\ \frac{d}{dx}[\psi(x, y(x))] = \psi_x \frac{dx}{dx} + \psi_y \frac{dy}{dx} \\ \qquad = \psi_x + \psi_y \frac{dy}{dx} \end{cases}$$

$$\hookrightarrow \Rightarrow \frac{d}{dx}\psi(x, y(x)) = 0$$

$$\Rightarrow \psi(x, y(x)) = C$$

② Implicitly defines $y = y(x)$

Condition: $\psi_x = M \Rightarrow (\psi_x)_y = M_y$

$\psi_y = N \Rightarrow (\psi_y)_x = N_x$

①, But $\psi_{xy} = \psi_{yx}$ for $\psi$ smooth

New Condition: — ①

✓✓ $M_y = N_x$ is sufficient for solvability

$M_y = N_x \Rightarrow$ $M(x,y) + N(x,y)\frac{dy}{dx} = 0$ is called exact

We call ② solve for all exact diff. eq.'s

ⓐ $\psi_x = M(x,y) \Rightarrow \psi = \int M(x,y)\,dx + f(y)$

ⓑ $\psi_y = N(x,y)$

→ (空二)

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(#) Sec. 2.6

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0 \quad (\text{eq. (1)})$$

Sec. 2.2

$$M(x) + N(y)\frac{dy}{dx} = 0 \quad \text{Separable}$$

[*] is solvable $\Leftrightarrow$ [*] is exact

$\Leftrightarrow M_y = N_x$

If $\exists \psi$, s.t. $\begin{cases} \psi_x = M & (\text{eq. (2)}) \\ \psi_y = N & (\text{eq. (3)}) \end{cases}$

$\Rightarrow$ then $(\psi_x)_y = M_y = N_x = (\psi_y)_x$

① $M + N\frac{dy}{dx} = 0 \Leftrightarrow \psi_x + \psi_y\frac{dy}{dx} = 0 \Leftrightarrow \frac{d}{dx}\psi(x, y(x)) = 0$

$\rightarrow$ Then $\psi(x,y) = C$ is a sol.)

If $M_y = N_x \Rightarrow$ then eq. (2) $\rightarrow \psi(x,y) = \int M(x,y)\,dx + f(y)$

$\psi_y = \int M_y(x,y)\,dx + f'(y) \overset{(3)}{=} N(x,y)$

$\int M_y(x,y)\,dx + f'(y) = N$

② $\int N_x(x,y)\,dx + f'(y) = N$

$\Leftrightarrow N(x,y) - g(y) - f'(y) = N(x,y)$

$f'(y) = g(y) \Rightarrow f(y) = \int g(y)\,dy + C$

Thus $\psi(x,y) = \int M(x,y)\,dx + \int g(y)\,dy + C$

E.g.: Solve $\underbrace{y\cos x + 2xe^y}_{M} + (\sin x + x^2 e^y - 1)\frac{dy}{dx} = 0$

$\begin{cases} M_y = \cos x + 2xe^y \\ N_x = \cos x + 2xe^y \end{cases}$

$\Rightarrow M_y = N_x$

$\rightarrow \psi_x = M = y\cos x + 2xe^y$

$\Rightarrow \psi(x,y) = y\sin x + x^2 e^y + f(y)$

$\rightarrow \psi_y = N = \sin x + x^2 e^y - 1$

$\begin{cases} \psi_y = \sin x + x^2 e^y + f'(y) \\ \rightarrow f'(y) = -1 \Rightarrow f(y) = -y + C \end{cases}$

$\Rightarrow$ The sol. is $\psi(x,y) = \cdots - y + C$

$\Rightarrow$ ? $\begin{cases} F(x,y) = 0 \\ y = \varphi(x) \\ F_y(x_0,y_0) \neq 0 \end{cases}$ ? $(x_0,y_0)$

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0$$

$$\Leftrightarrow \mu(x,y)M(x,y) + \mu(x,y)N(x,y)\frac{dy}{dx} = 0$$

$\underline{N_x \neq M_y}$

But (可能) $(\mu M)_y = (\mu N)_x$

$$\mu_y M + \mu M_y = \mu_x N + \mu N_x$$

If $\mu = \mu(x) \rightarrow \mu_y = 0,\ \mu_x = \mu'$

$$\mu M_y = \mu' N + \mu N_x$$

If $\frac{M_y - N_x}{N}$ is a function of $x \rightarrow$ 记为 $f(x) = \frac{M_y - N_x}{N}$

① $\Rightarrow$ then $\mu f = \frac{d\mu}{dx} \Rightarrow \int \frac{d\mu}{\mu} = \int f(x)\,dx$

$$\Rightarrow \mu(x) = e^{\int f(x)\,dx}$$

② $\frac{\mu(M_y - N_x)}{N} = \mu'$

If $\mu = \mu(y) \rightarrow \mu_y = \mu',\ \mu_x = 0$

$$\mu' M + \mu M_y = \mu N_x$$

$$\Rightarrow \mu' = \frac{N_x - M_y}{M}\mu(y)$$

If $\frac{N_x - M_y}{M}$ is a func. of $y \rightarrow$ 记为 $g(y)$

$\Rightarrow$ then $\mu(y) = e^{\int g(y)\,dy}$

E.g.: Solve $\underbrace{(3xy+y^2)}_{M}+\underbrace{(x^2+xy)}_{N}\frac{dy}{dx}=0$

$M_y = 3x+2y$

$N_x = 2x+y$

$$\frac{M_y - N_x}{N} = \frac{3x+2y-(2x+y)}{x^2+xy}$$

$$= \frac{\cancel{x+y}}{x\cancel{(x+y)}} = \frac{1}{x}$$

$$\mu' = \frac{1}{x}\mu$$

$$\Rightarrow \int \frac{d\mu}{\mu} = \int \frac{dx}{x}$$

$$\Rightarrow \ln|\mu| = \ln|x| \Rightarrow \mu = x$$

